

Partial Least Squares-regression (PLS-regression) In Chemometrics

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Abstract: Partial least squares regression (PLS- regression) is a statistical method. instead of finding hyperplanes of minimum variance between the response and independent variables, it finds a linear regression model by projecting the predicted variables and the observable variables to a new space regression is effectively used in process modeling and monitoring to deal with a large number of variables with collinearity. usage in PLS regression and cross-validation. As a natural extension, the recursive algorithm is extended to dynamic modeling and nonlinear modeling. The analysis of mixture data is a common problem in industrial research and development, particularly in chemical and related industries, e.g. pharmaceuticals, cosmetics, oil, and biotechnology. limitation of multiple regression with data in constrained regions. For the analysis of mixture data, partial least squares (PLS) has been found to be practical. In particular when both mixture and process variables are involved, it offers a flexible and simple approach which works well in practice.

Index terms: Partial Least Squares, PLS- regression, Chemometrics, Analytical Chemistry.

I. INTRODUCTION

It is almost 47 years since chemometrics started in its modern form. As then and now, efforts have been made to reach an acceptable definition of chemometrics often something along the lines of “how to get chemically relevant information out of measured chemical data, how to represent and display this information, and how to get such information into data”[1] Hence, chemometrics can be considered as an area that took off with the advent of scientific computing, especially with the development of computerized laboratory based instrumentation.[2] However, over the years, other applied informatics and computationally based disciplines have overtaken chemometrics. Bioinformatics, still in its infancy before the human genome project; quantum structure–activity relationships, not yet joined at the hip to quantum chemistry; and chemoinformatics have all overtaken chemometrics as accepted and formal core scientific disciplines.

Chemometrics is the science of extracting information from chemical systems by data-driven means. Chemometrics is inherently interdisciplinary, using methods frequently employed in core data-analytic disciplines such as multivariate statistics, applied mathematics, and computer science, in order to address problems in chemistry, biochemistry, medicine, biology and chemical engineering. In this way, it mirrors other interdisciplinary fields, such as psychometrics and econometrics.

Chemometrics is applied to solve both descriptive and predictive problems in experimental natural sciences, especially in chemistry. In descriptive applications, properties of chemical systems are modeled with the intent of learning the underlying relationships and structure of the system (i.e., model

understanding and identification). In predictive applications, properties of chemical systems are modeled with the intent of predicting new properties or behavior of interest. In both cases, the datasets can be small but are often very large and highly complex, involving hundreds to thousands of variables, and hundreds to thousands of cases or observations.

Chemometric techniques are particularly heavily used in analytical chemistry and metabolomics, and the development of improved chemometric methods of analysis also continues to advance the state of the art in analytical instrumentation and methodology. It is an application-driven discipline, and thus while the standard chemometric methodologies are very widely used industrially, academic groups are dedicated to the continued development of chemometric theory, method and application development.

Although one could argue that even the earliest analytical experiments in chemistry involved a form of chemometrics, the field is generally recognized to have emerged in the 1970s as computers became increasingly exploited for scientific investigation. The term ‘chemometrics’ was coined by Svante Wold in a grant application 1971, and the International Chemometrics Society was formed shortly thereafter by Svante Wold and Bruce Kowalski, two pioneers in the field. Wold was a professor of organic chemistry at Umeå University, Sweden, and Kowalski was a professor of analytical chemistry at University of Washington, Seattle.

Many early applications involved multivariate classification, numerous quantitative predictive applications followed, and by the late 1970s and early 1980s a wide variety of data- and computer-driven chemical analyses were occurring.

Multivariate analysis was a critical facet even in the earliest applications of chemometrics. The data

resulting from infrared and UV/visible spectroscopy are often easily numbering in the thousands of measurements per sample. Mass spectrometry, nuclear magnetic resonance, atomic emission/absorption and chromatography experiments are also all by nature highly multivariate. The structure of these data was found to be conducive to using techniques such as partial least-squares (PLS). This is primarily because, while the datasets may be highly multivariate there is strong and often linear low-rank structure present. PLS have been shown over time very effective at empirically modeling the more chemically interesting low-rank structure, exploiting the interrelationships or 'latent variables' in the data, and providing alternative compact coordinate systems for further numerical analysis such as regression, clustering, and pattern recognition. Partial least squares in particular was heavily used in chemometric applications for many years before it began to find regular use in other fields.

Thing for sure is that pattern recognition is fast becoming one of the major, if not the major, application of chemometrics. So many applications ranging from heritage studies to metabolomics to forensics require pattern recognition.

A common problem in the chemical, pharmaceutical, and similar industries is the analysis of mixture data. With mixture data, the factors are expressed as proportions of the total amount. Since the pioneering work of Wold (1966), partial least squares (PLS) regression has been widely applied in chemometrics (Lindberg et al., 1983; Wold et al., 1984; Geladi and Kowalski, 1986; Fuller et al., 1988; and Haaland and Thomas, 1988; Martin and Naes).

PLS regression is usually biased with reduced variance; as a result, the overall mean square error is minimized (Hoskuldsson, 1988).

In most of the PLS applications to date, the PLS regression is a batch-wise modeling approach. In other words, the data are collected and stored in a computer, then the PLS regression is carried out on the whole batch of data. While the batch type PLS circumvents the collinearity problem, it has limitations in the following situations. First, it is difficult to update a PLS model online using newly available data. While one could rebuild a new model based on merging the new data and old data, it is computationally inefficient because the old data are modeled repeatedly. Second, in the case of large data sets with many variables and data samples, which is often encountered in process data analysis, the batch PLS algorithm may run out of computer memory for a given computing platform. Third, the typical crossvalidation procedure involves time-consuming and repetitive calculation by leaving out a subset of data and modeling on the remaining subsets. It is desirable to improve the computation efficiency by reusing the previous calculation in the procedure [3].

The PLS algorithm is employed in partial least squares path modeling,[4][5] a method of modeling a "causal" network of latent variables (causes cannot be determined without experimental or quasi-experimental methods, but one typically bases a latent variable model on the prior theoretical assumption that latent variables cause manifestations in their measured indicators). This technique is a form of structural equation modeling, distinguished from the classical method by being component-based rather than covariance-based.[6].

Partial least squares was introduced by the Swedish statistician Herman Wold, who then developed it with his son, Svante Wold. An alternative term for PLS (and more correct according to Svante Wold[7]) is projection to latent structures, but the term partial least squares is still dominant in many areas. Although the original applications were in the social sciences, PLS regression is today most widely used in chemometrics and related areas. It is also used in bioinformatics, sensometrics, neuroscience and anthropology. In contrast, PLS path modeling is most often used in social sciences, econometrics, marketing and strategic management.

Partial least squares (PLS) has been found to be practical for the analysis of mixture data. In particular, when both mixture and process factors are involved, it offers a flexible and simple approach which works well in practice[8].

II. Partial Least Squares- regression (PLS-regression) Model

Any model needs to be validated before it is used for understanding or for predicting new events such as the biological activity of new compounds or the yield and impurities at other process conditions.

PLS is used to find the fundamental relations between two matrices (X and Y), i.e. a latent variable approach to modeling the covariance structures in these two spaces. A PLS model will try to find the multidimensional direction in the X space that explains the maximum multidimensional variance direction in the Y space. PLS regression is particularly suited when the matrix of predictors has more variables than observations, and when there is multicollinearity among X values. By contrast, standard regression will fail in these cases (unless it is regularized). In this section we briefly discuss the traditional batch PLS algorithm in order to derive the recursive PLS algorithm. Given a pair of input and output data matrices X and Y and assuming they are linearly related by

$$Y=XC+V \quad (1)$$

where V and C are noise and coefficient matrices, respectively, the PLS regression builds a linear model by decomposing matrices X and Y into bilinear terms,

The general underlying model of multivariate PLS is

$$X = TP^T + E \quad (2)$$

$$Y = UQ^T + F \quad (3)$$

where X is an $n \times m$ matrix of predictors, Y is an $n \times p$ matrix of responses; T and U are $n \times 1$ matrices that are, respectively, projections of X (the *X score, component* or *factor* matrix) and projections of Y (the *Y scores*); P and Q are, respectively, $m \times 1$ and $p \times 1$ orthogonal *loading* matrices; and matrices E and F are the error terms, assumed to be independent and identically distributed random normal variables. The decompositions of X and Y are made so as to maximise the covariance between T and U .

The matrix X refers to the predictor variables and their squares or/and cross terms if these have been added. Active x variables participating in the model are often referred to as terms. When squares and cross terms are added to the X matrix this corresponds to fitting quadratic models. The PLS model consists of a simultaneous projection of both the X and Y spaces on a low-dimensional hyperplane. The coordinates of the points on this hyperplane constitute the elements of the matrix T . This analysis has the following two objectives: (1) to approximate well the X and Y spaces; (2) to maximize the correlation between X and Y .

PLS contains the multiple regression solution as a special case, i.e. with one response and as many PLS components (A) as there are non-zero singular values of X ; the PLS model gives identical predictions and response surfaces as Cox or Scheffe models. The PLS coefficients is then identical to the Cox's, when expressed relative to the same reference mixture. In PLS modelling, we assume that the investigated system or process actually is influenced by just a few underlying variables.

III. Least Squares- regression (PLS- regression) Algorithms

The algorithms for calculating the PLSR model are mainly of technical interest, we here just point out that there are several variants developed for different shapes of the data.[9]

A number of variants of PLS exist for estimating the factor and loading matrices T, U, P and Q . Most of them construct estimates of the linear regression between X and Y as $Y = XB + B_0$. Some PLS algorithms are only appropriate for the case where Y is a column vector, while others deal with the general case of a matrix Y . Algorithms also differ on whether they estimate the factor matrix T as an orthogonal, an orthonormal matrix or not.[10][11][12][13][14] The final prediction will be the same for all these varieties of PLS, but the components will differ.

IV. Homogeneity Partial Least Squares- regression (PLS- regression)

Any data analysis is based on an assumption of homogeneity. This means that the investigated system or process must be in a similar state throughout all the investigation, and the mechanism of influence of X on Y must be the same. This, in turn, corresponds to having some limits on the variability and diversity of X and Y .

Hence, it is essential that the analysis provides diagnostics about how well these assumptions indeed are fulfilled. Much of the recent progress in applied statistics has concerned diagnostics [15], and many of these diagnostics can be used also in PLSR modeling as discussed below. PLSR also provides additional diagnostics beyond those of regression-like methods, particularly those based on the modelling of X (score and loading plots and X -residuals).

In the first example, the first PLSR analysis indicated that the data set was inhomogeneous—three aromatic amino acids (AA's). are indicated to have a different type of effect than the others on the modeled property. This type of information is difficult to obtain in ordinary regression modelling, where only large residuals in Y provide diagnostics about inhomogeneities.[6]

V. Non-linear Partial Least Squares- regression (PLS- regression)

For non-linear situations, simple solutions have been published by Höskuldsson [15], and Berglund and Wold [16]. Another approach based on transforming selected X -variables or X -scores to qualitative variables coded as sets of dummy variables, the so-called GIFI approach [17][18], is described elsewhere in this volume [19].

CONCLUSION

Partial least squares regression (PLS- regression) is a statistical method. instead of finding hyperplanes of minimum variance between the response and independent variables, it finds a linear regression model by projecting the predicted variables and the observable variables to a new space regression is effectively used in process modeling and monitoring to deal with a large number of variables with collinearity.

PLSR provides an approach to the quantitative modelling of the often complicated relationships between predictors, X , and responses, Y , that with complex problems often is more realistic than MLR including stepwise selection variants. This because the assumptions underlying PLS—correlations among the X 's, noise in X , model errors—are more realistic

than the MLR assumptions of independent and error free X's.

The ability of PLSR to analyze *profiles* of responses, makes it easier to device response measurements that are relevant to the stated objective of the investigation; it is easier to capture the behavior of a complicated system by a battery of measurements than by a single variable.

We feel that the flexibility of the PLS-approach, its graphical orientation, and its inherent ability to handle incomplete and noisy data with many variables $\bar{Z}$ and observations. makes PLS a simple but powerful approach for the analysis of data of complicated problems.

As for the model estimation, if one uses projection methods that do not assume independence of the factors, such as PLS, there is no need to treat or think of mixture variables as different from ordinary process variables. Rather, one should think of all relevant variables as factors, and express their constraint for the problem at hand. Then one selects a model that corresponds to one's objectives, with the appropriate metric (transformations), to obtain the best fit, predictions and interpretability.

With PLS, as illustrated by the examples, the analysis of mixture data is simple, straightforward, and works well in practice.

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